

Zero-Shot Video Anomaly Detection using Vision-Language Models with Domain-Adaptive Verbalized Prompting



Zero-Shot
No Training Data



Vision-Language
CLIP-based Embeddings



Domain-Adaptive
Verbalized Prompting

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Guide: **Prof. Dr. Pranesh Das**

The Problem & The Challenge

Why zero-shot anomaly detection is the missing link in conventional vision systems.



Data Dependency

Traditional AD requires expensive training data **per category**, limiting scalability across diverse domains.



Lack of Explainability

Outputs only numerical scores — **never explains** what is wrong, hindering root cause analysis.



Temporal Blindness

No temporal awareness for video streams, treating every frame as a static, isolated event.



The Critical Gap

No existing method is zero-shot, explainable, and temporal **simultaneously**.

Our Approach bridges this gap using CLIP's vision-language space for a **zero-shot, explainable, and scalable** path forward.

Each stage contributes to a lightweight inference pipeline that can extend from static images to video.

Literature: PatchCore & WinCLIP

Foundational image anomaly detection methods — our primary baselines



PatchCore • CVPR 2022

- Patch-level features in a **memory bank**
- Anomaly = max distance from normal patches
- **99.1% AUROC** on MVTec AD
- Requires category-specific training; no explanations



WinCLIP • CVPR 2023

- First CLIP-based **zero-shot** AD method
- **Prompt Ensembles** + sliding window
- **91.8% AUROC** zero-shot on MVTec AD
- Hand-crafted prompts; no defect explanations

 Our baseline: **88.5% zero-shot AUROC** using prompt ensembles, no sliding window — kept simple for video extensibility.



ADVANCED VLMS

RESEARCH GAP

AnomalyCLIP, AnomalyGPT & MMAD

Advanced VLM approaches — and why a research gap persists



AnomalyCLIP • ICLR 2024

- Learnable **object-agnostic prompts**
- **~94% AUROC** on MVTec AD
- Better cross-domain generalisation
- Requires fine-tuning — **not zero-shot**
- No video or temporal capability



AnomalyGPT • AAI 2024

- First MLLM for industrial AD
- **94.1% AUROC** with 1-shot
- Natural language **explanations** of defects
- Requires fine-tuning — not zero-shot
- No temporal support



MMAD • ICLR 2025

- Benchmark: **7 AD subtasks**
- Tests GPT-4o, Claude, LLaVA
- GPT-4o: **moderate accuracy only**
- Prompt design critical for AD
- MLLMs alone are **insufficient**



Research gap validated: No existing method is simultaneously zero-shot, explainable, and temporally aware.

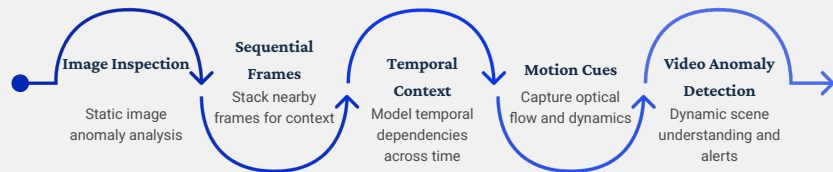
Pivot to Video AD

Why we extended from image anomaly detection to video anomaly detection



Why extend to video?

- Image AD: static, controlled environments
- Video AD: dynamic scenes with **temporal context**
- Professor's guidance: *"Look into domain adaptability beyond static images"*



[OPEN-VOCABULARY](#)[VIDEO AD](#)[SURVEY](#)

OVVAD & VAD Survey: Shaping Our Direction

CVPR 2024 + ACM 2025: Zero-shot framework and comprehensive survey evidence.



OVVAD Architecture

- **Frozen CLIP:** Fixed ViT-B/16 features
- **Temporal adapter:** Learns frame sequence cues
- **Text injection:** Normal/anomaly descriptions
- **Anomaly synthesis:** Boosts robustness

Performance: 86.40% AUC on UCF-Crime



Why the Survey Matters

- **400+ methods:** Provides a broad VAD landscape map
- **Key gaps:** Generalization, time modeling, label-heavy
- **Open-vocab VAD:** Handles unseen anomalies via language
- **Strategic Direction:** Supports zero-shot, language-aware DA-ZVAD

LAVAD & VERA: Training-Free Video AD

Two CVPR papers that complete our architectural puzzle



LAVAD: Caption-then-reason

- Frames → **BLIP-2 Captions** → LLM Scoring
- LLM reasons about **temporal events**
- Dynamically adjusts normality baseline
- Outperforms all unsupervised VAD methods

Limitations: Noisy captions; high inference cost



VERA: Domain Prompting

- *"A dimly lit warehouse with conveyor belts"*
- Cross-modal refinement reduces noise
- **Zero training required**; adaptive via text
- Explainable outputs; cross-domain robust

Running person: Normal in an arena, **anomalous in a warehouse** — domain context matters.

Comparative Analysis

No existing method satisfies all five desiderata — **DA-ZVAD** targets this gap.

What matters most: zero-shot, training-free, explainable, temporal, and domain-adaptive performance.

Method	Venue	Zero-Shot	Training-Free	Explainable	Temporal	Domain-Adaptive
PatchCore	CVPR '22	X	X	X	X	X
WinCLIP	CVPR '23	✓	✓	X	X	X
AnomalyCLIP	ICLR '24	✓	X	X	X	X
Anomaly GPT	AAAI '24	X	X	✓	X	X
OVVAD	CVPR '24	✓	X	X	✓	Partial
LAVAD	CVPR '24	✓	✓	Partial	✓	X
VERA	CVPR '25	✓	✓	✓	✓	✓
DA-ZVAD (Ours)	Proposed	✓	✓	✓	✓	✓

Key insight: DA-ZVAD is the only proposed method satisfying all five desiderata simultaneously — zero-shot, training-free, explainable, temporal, and domain-adaptive.

Challenges In Existing Study

1. Training Dependency

PatchCore, AnomalyGPT, OVVD require costly domain-specific data and retraining.

2. Closed-Vocabulary

Traditional methods fail silently on novel, unseen anomaly categories at test time.

3. No Explainability

Numerical scores lack reasoning, making outputs unusable for industrial compliance.

4. Domain Rigidity

Normality is context-dependent; fixed methods cannot adapt across gym vs. hospital.

5. No Temporal Reasoning

Frame-based scoring misses progressive events like walking → running → attacking.

6. Caption Hallucination

Noisy VLM captions in methods like LAVAD lead to unreliable reasoning and false positives.

Problem Statement & Objectives

"Design and Development of a Zero-Shot Video Anomaly Detection System using Vision-Language Models with Domain-Adaptive Verbalized Prompting for Cross-Domain Surveillance and Industrial Monitoring"

Objectives

- Develop a **training-free** VAD pipeline using pre-trained Vision-Language Models
- Enable **domain adaptation** via natural language context descriptions
- Provide **natural language explanations** for every detection
- Achieve **open-vocabulary** detection of unseen anomaly types
- Evaluate across **cross-domain benchmarks** (UCF, XD, MVTec)

Key Challenges

- Costly domain-specific training – poor scalability
- No explainability – output scores only, no reasoning
- Domain rigidity – context-dependent normality treated as fixed
- No temporal reasoning – missing progressive anomalous events
- Caption hallucination – noisy VLM descriptions causing false positives

FORMULATION

Mathematical Mapping:

- Input: Video $V = \{f_1, f_2, \dots, f_n\}$
- Output: Anomaly score $S(f_i) \in [0, 1]$
- Domain context: $C = \text{"A hospital hall"}$

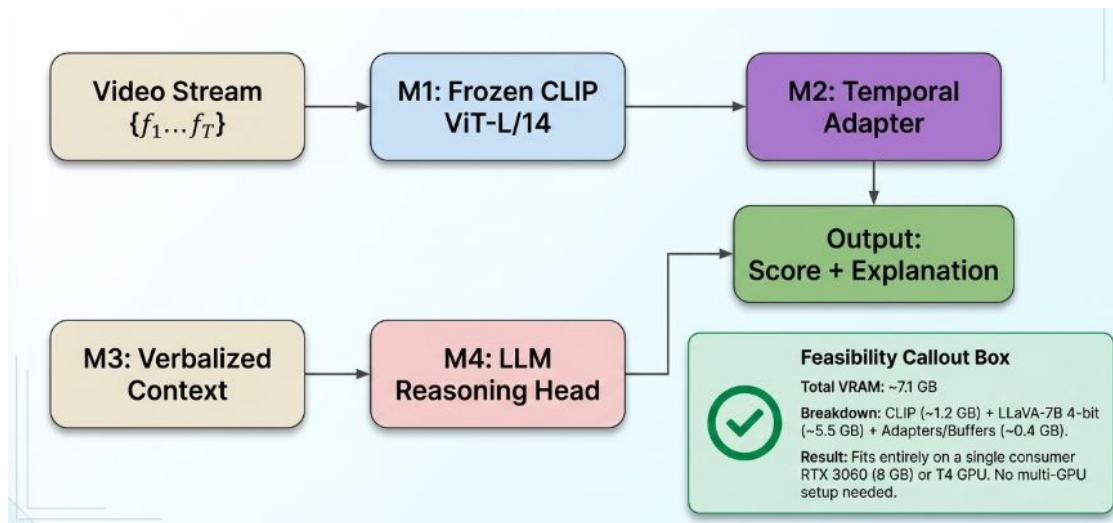
Key Idea: $S(f_i | C)$ – detection depends entirely on where the camera is deployed.

Example: "Running Person"

- Context = "Gym" → Score = **0.1 (Normal)**
- Context = "Hospital" → Score = **0.9 (Anomaly)**

DA-ZVAD — Proposed Architecture

A clean, end-to-end system that turns video input into an anomaly score and a natural-language explanation.



- ✓ **Feasibility:** Total VRAM ~7.1 GB (CLIP ViT-L/14 ~1.2 GB + LLaVA-7B 4-bit ~5.5 GB + Temporal Adapter ~0.2 GB + Frame Buffer ~0.2 GB).
Runs on a single RTX 3060 (8 GB) or T4. No multi-GPU setup needed.

CLIP Baseline — Results on MVTec AD

15 categories · 1,725 test images · Zero training data

88.5%

Mean AUROC

Zero-shot, no training

90.7%

F1 Score

Strong detection precision

135.8s

Inference Time

On T4 GPU, full dataset

Gap vs WinCLIP (91.8%): attributed to omission of multi-scale sliding window — a deliberate trade-off for video extensibility.

Category Breakdown

Textures — 99.5% mean

*Carpet 99.2% · Grid 99.7% · Leather **100%** · Tile 98.7% · Wood **100%***

✓ Global appearance captured well by CLIP

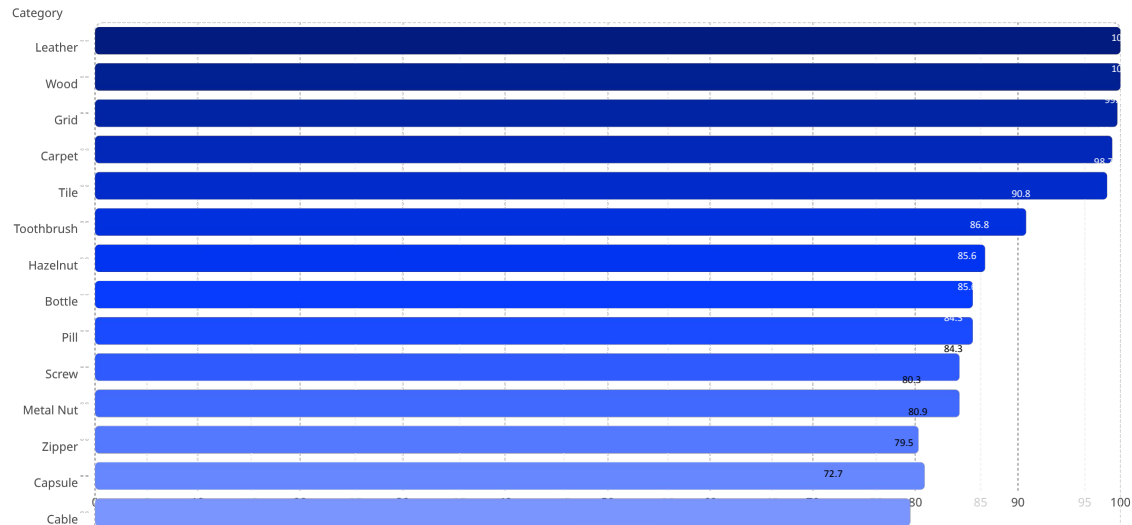
Objects — 83.1% mean

Bottle 85.6% · Cable 79.5% · Capsule 80.9% · Transistor 72.7% · Zipper 80.3%

⚠ Localised defects need stronger spatial reasoning

AUROC by Category

Visual performance breakdown across all 15 MVtec AD categories — textures consistently outperform objects.



Key pattern: texture categories are nearly perfect, while object categories are more sensitive to localized and structural defects.

Textures lead

Leather, wood, grid, carpet, and tile cluster at the top.

Objects lag

Transistor, cable, capsule, and zipper show the toughest cases.

Next phase

Improve spatial reasoning for fine-grained defect localization.

Texture categories (leather, wood, grid) achieve near-perfect scores. Object categories with localised or structural defects — transistor, cable, capsule — highlight the need for spatial reasoning enhancements in the next phase.

Video Extension — Per-Frame CLIP Scoring

Simulating conveyor belt quality inspection with pseudo-video frames

0.307

Normal Score

0.894

Anomalous Score

0.587

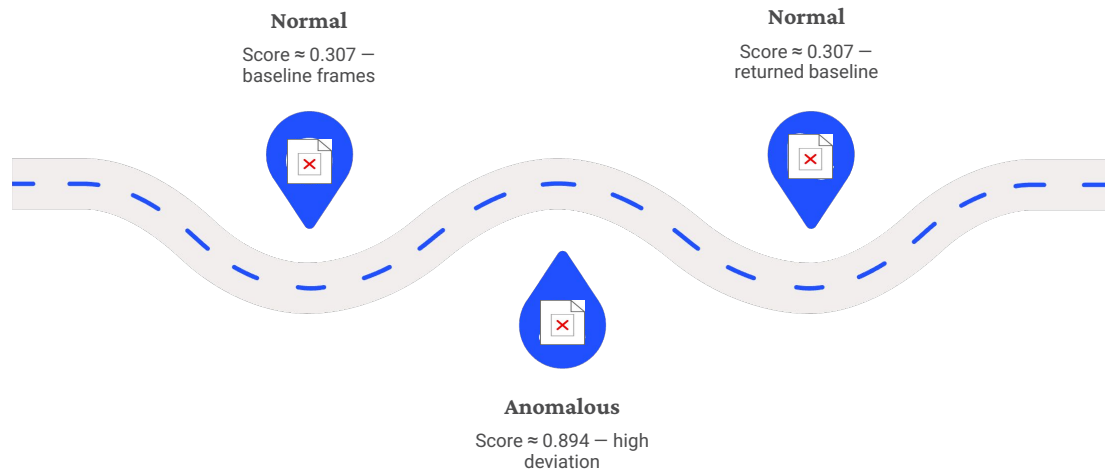
Score Gap

Experimental Setup

- MVTec bottle images sequenced as **pseudo-video frames**
- 30 frames: Normal $\times 10 \rightarrow$ Anomalous $\times 10 \rightarrow$ Normal $\times 10$
- Each frame scored **independently** by CLIP
- Simulates real-time conveyor belt inspection

Frame Sequence — Score

Pattern



Clean separation validates CLIP for video: A 0.587 score gap between normal and anomalous frames confirms CLIP's discriminative power as a foundation for the DA-ZVAD video pipeline.

Roadmap & Expected Contributions

1

Semester 2 — *Done*

Literature survey (15 papers), CLIP baseline 88.5% AUROC, video proof-of-concept, DA-ZVAD design

2

Semester 3 — *Jul–Nov*

UCF-Crime setup, OVVD temporal adapter, LAVAD caption-then-reason, VERA verbalized prompting

3

Semester 4 — *Jan–May*

Full DA-ZVAD integration, cross-domain evaluation, ablation studies, thesis writing

Expected Contributions

1. Systematic prompt engineering study
2. Hybrid VLM + LLM video AD pipeline
3. Cross-domain benchmark evaluation
4. Domain-adaptive verbalized prompting

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Thank You

Questions & Discussion

Prakash Kumar Sarangi | M251250CS | NIT Calicut | github.com/prakash-nitc/multimodal-anomaly-detection